

Surface Area Using Nets

Lesson 10-3

Name: _____

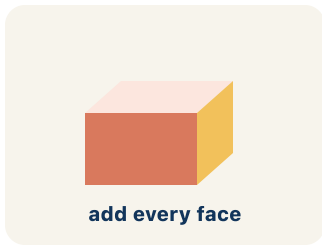
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



*A box $6 \times 4 \times 2$: $SA = 2(24) + 2(12) + 2(8) = 88 \text{ in}^2$
— like wrapping paper needed*

Surface area

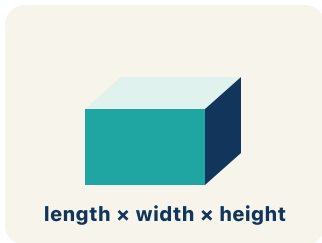
Write the definition:



*Cut along the edges of a cereal box and unfold it
flat — you see all 6 faces*

Net

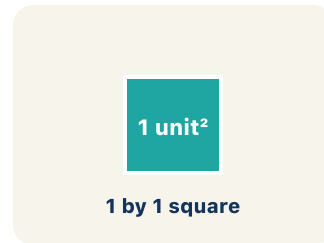
Write the definition:



*A rectangular prism has 6 faces: top, bottom, front,
back, left, right*

Face

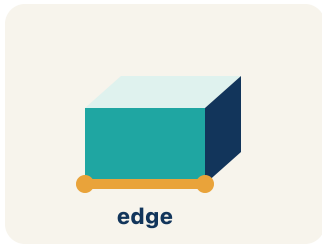
Write the definition:



*A rectangle drawn on paper is 2D — it has length
and width but no depth*

Two-dimensional

Write the definition:



A cube has 12 edges — the lines where two flat faces connect

Edge

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can find the surface area of a solid by using its net.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Surface area

Net

Face

Two-dimensional

Edge

1 The total area of all the faces of a solid is its _____.

2 A flat pattern that folds up into a solid is a _____.

3 A flat surface of a solid figure is a _____.

4 A flat shape with length and width but no thickness is _____.

5 A line segment where two faces meet is an _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

I do — watch

Follow each step as your teacher solves it.

Problem: A rectangular prism has $l = 6$ in, $w = 4$ in, $h = 2$ in. What is its surface area?

A. 88 in^2

B. 48 in^2

C. 88 in^3

D. 44 in^2

Step 1 $SA = 2(6 \times 4) + 2(6 \times 2) + 2(4 \times 2) = 48 + 24 + 16 = 88 \text{ in}^2$.

Answer: A. 88 in^2

 We do — together

Solve this one with your class using the same steps.

Problem: How many faces does a rectangular prism have?

- A. 6
- B. 4
- C. 8
- D. 12

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Surface area is measured in which type of unit?

- A. Square units (in^2 , cm^2 , ft^2)
- B. Cubic units (in^3 , cm^3 , ft^3)
- C. Linear units (in, cm, ft)
- D. No units needed

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Your team unfolds a rectangular time capsule box into a flat net. How many faces (flat sides) will the net show in all?

- A. 6
- B. 4
- C. 5
- D. 8

Show your work:

2. A time capsule box measures 2 in $\times$ 3 in $\times$ 4 in. Using its net, you add up all six faces. What is the total surface area?

- A. 52 in²
- B. 24 in²
- C. 26 in²
- D. 48 in²

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Two boxes have the same volume of 60 in^3 . Box A is $5 \times 4 \times 3$. Box B is $10 \times 6 \times 1$. Which box uses less wrapping paper (has less surface area)? Show your calculations and explain why a more cube-like shape might use less material.

Sentence starter: Box A: $SA = 2(\text{---}) + 2(\text{---}) + 2(\text{---}) = \text{---} \text{ in}^2$. Box B: $SA = 2(\text{---}) + 2(\text{---}) + 2(\text{---}) = \text{---} \text{ in}^2$. Box \text{---}.

Show your work:

Reflect — Exit Ticket

A rectangular prism has $l = 9 \text{ in}$, $w = 5 \text{ in}$, $h = 3 \text{ in}$. What is the surface area?

- A. 174 in^2
- B. 135 in^2
- C. 174 in^3
- D. 87 in^2

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Surface area
2. **Notes 2:** Net
3. **Notes 3:** Face
4. **Notes 4:** Two-dimensional
5. **Notes 5:** Edge
6. **We Try (notes):** A. 6 — *A rectangular prism has 6 faces: top, bottom, front, back, left, and right.*
7. **You Try (notes):** A. Square units (in^2 , cm^2 , ft^2) — *Surface area measures flat surfaces, so it uses square units. Volume uses cubic units.*
8. **Try It 1:** A. 6 — *A rectangular prism has 6 faces — top, bottom, front, back, left, and right — so its net shows 6 faces.*
9. **Try It 2:** A. 52 in^2 — *From the net: $SA = 2(2 \times 3) + 2(2 \times 4) + 2(3 \times 4) = 12 + 16 + 24 = 52 \text{ in}^2$.*
10. **Exit Ticket:** A. 174 in^2 — *$SA = 2(9 \times 5) + 2(9 \times 3) + 2(5 \times 3) = 90 + 54 + 30 = 174 \text{ in}^2$. Surface area uses square units (in^2).*